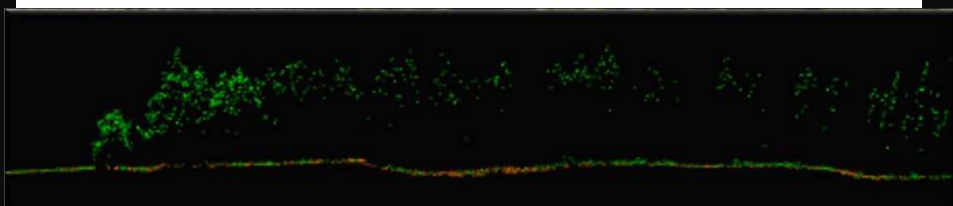
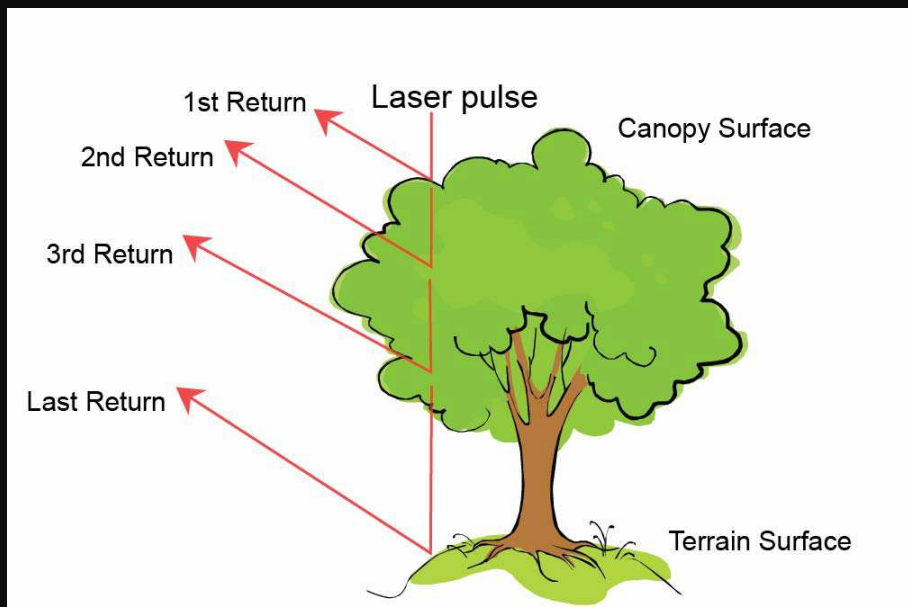


LiDAR

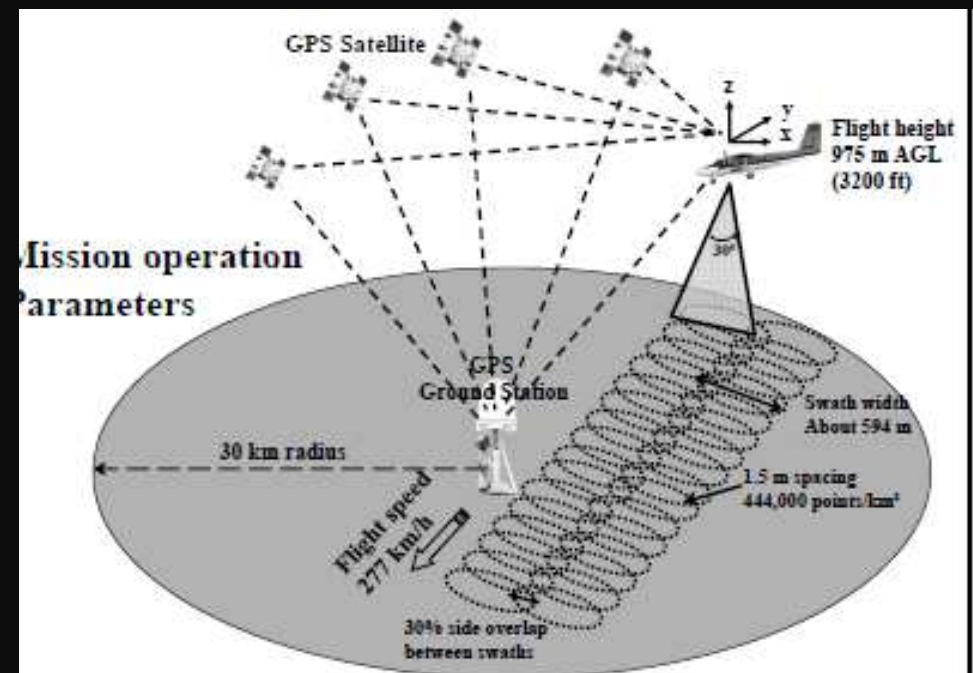
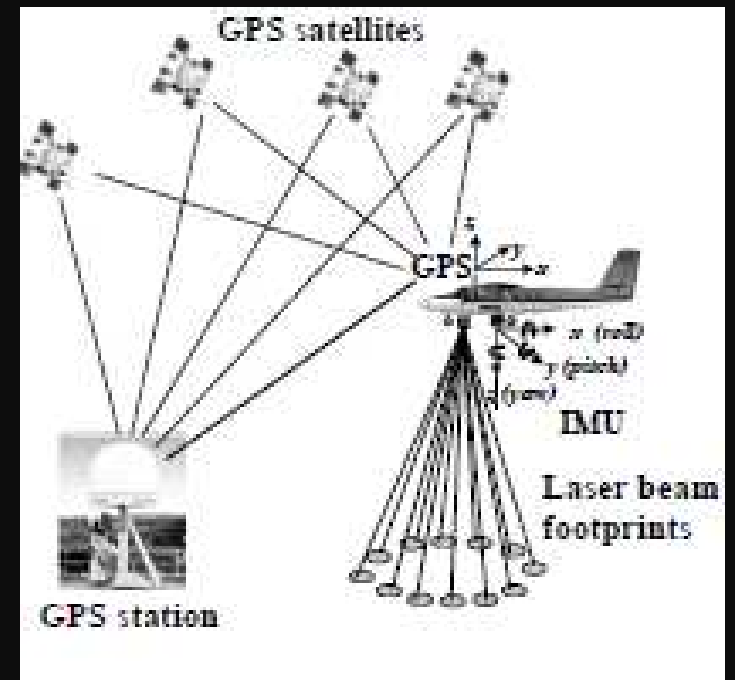
- *LiDAR = Light Detection And Ranging (Laser Radar)*
- *Uses an active sensor to emit energy (light) and detect returned energy*
- *Can be collected day or night*

How it works?



Lasers is 1064 nm – near infrared region

System Components



LiDAR Data

All returns

1st return

2nd return

3rd return

4th return

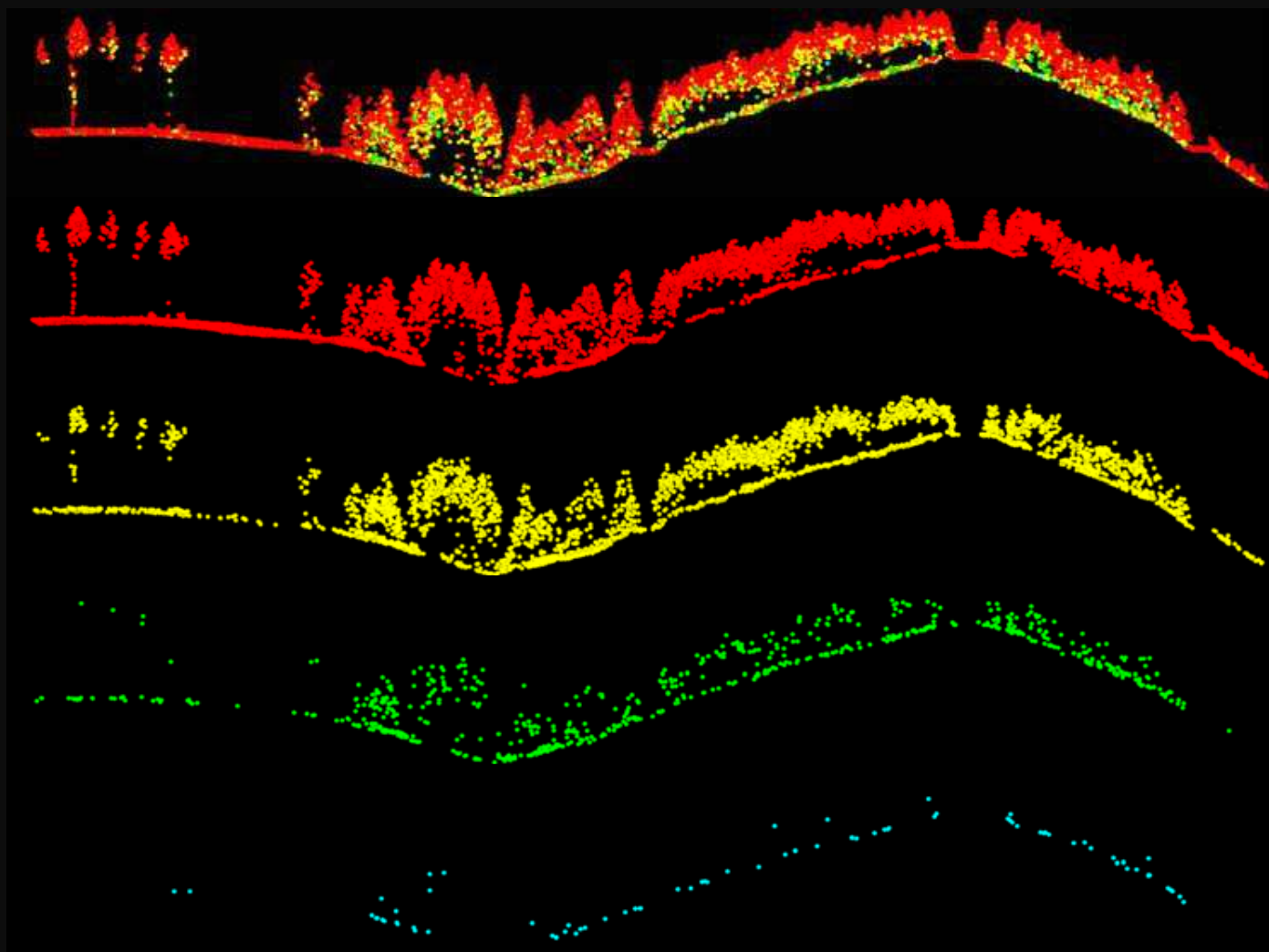
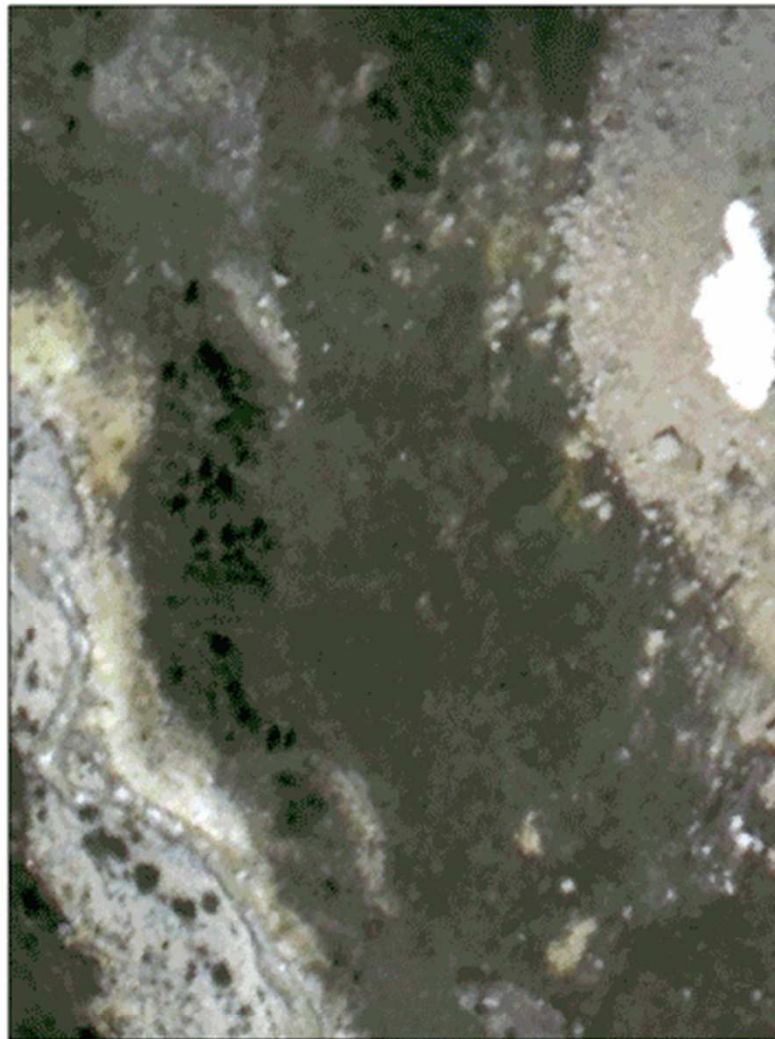
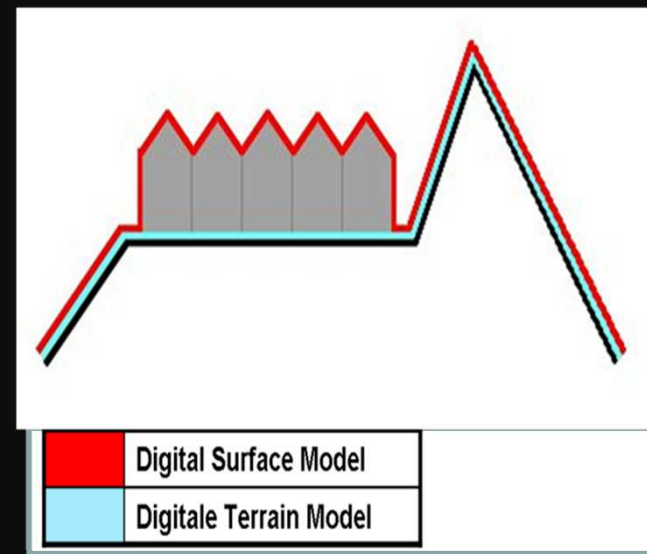
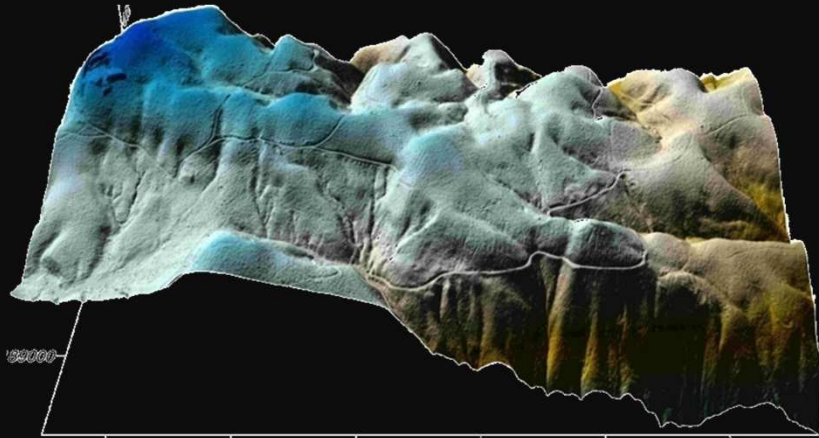


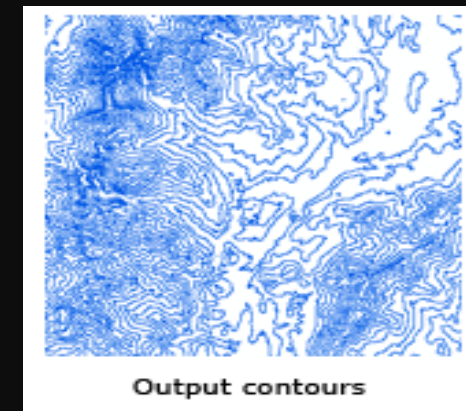
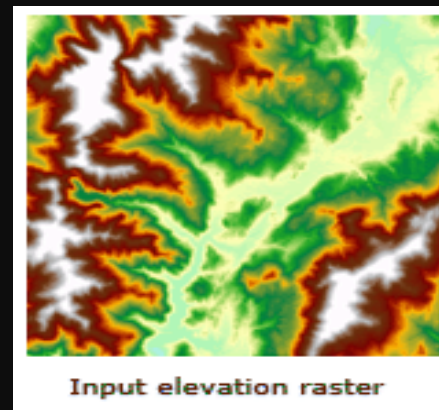
Image from Lidar Technology Overview, presentation by USGS, June 2007



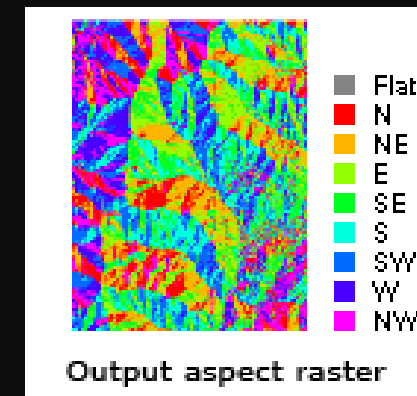
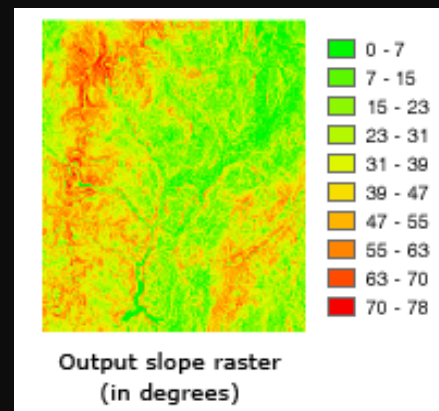
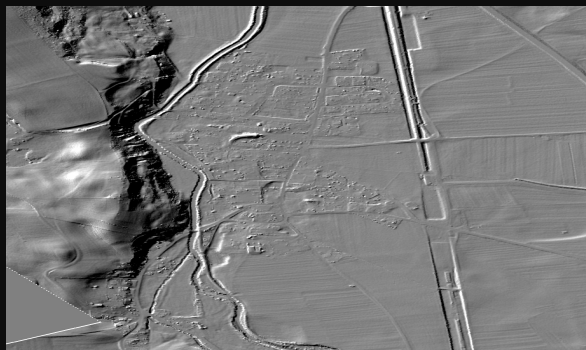
DEM



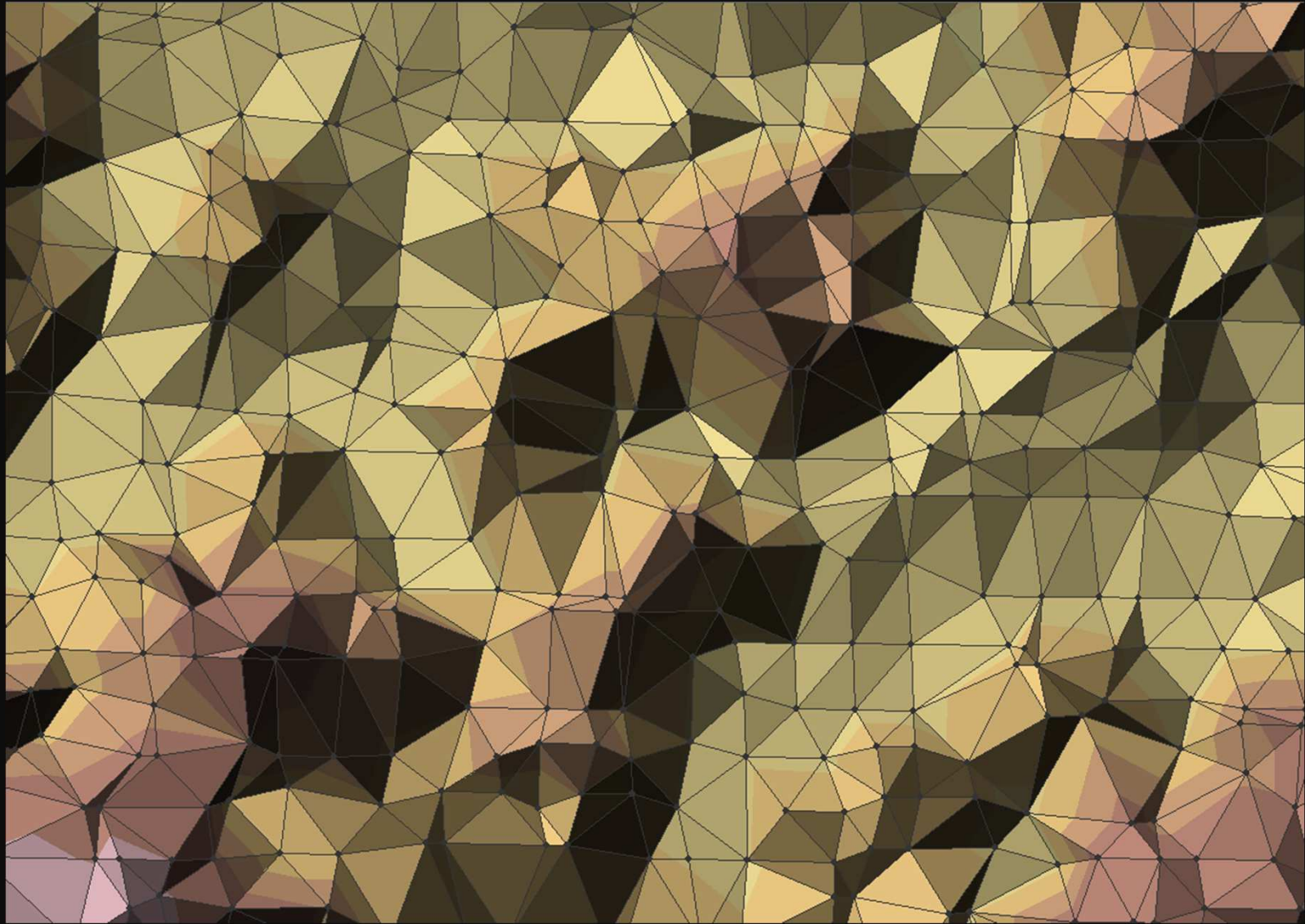
DSM



DTM



A TIN creating from LiDAR data



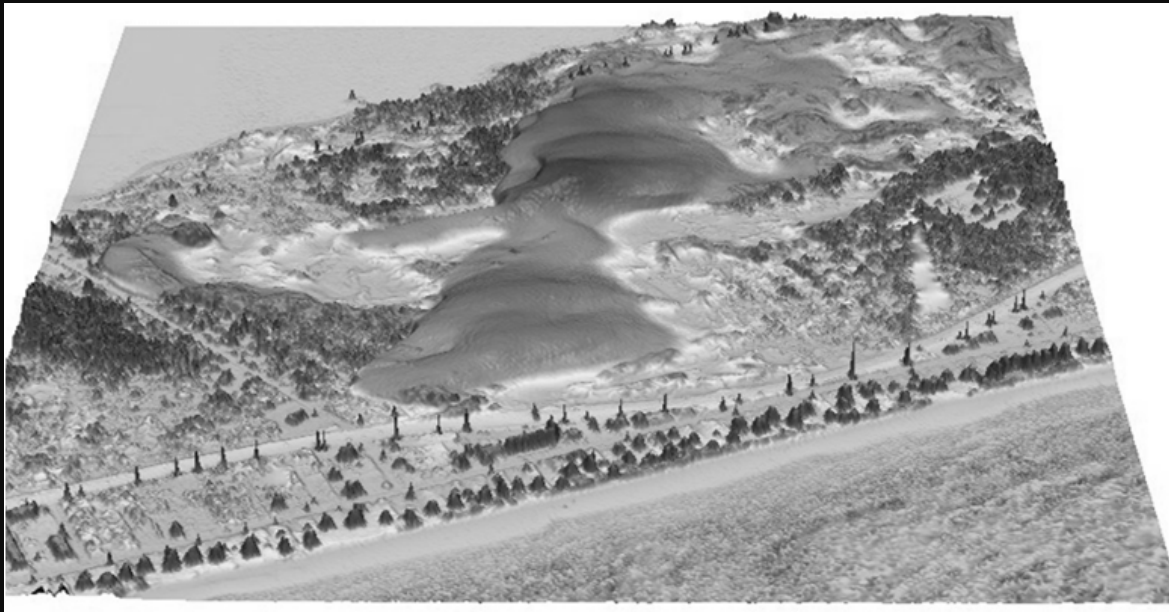
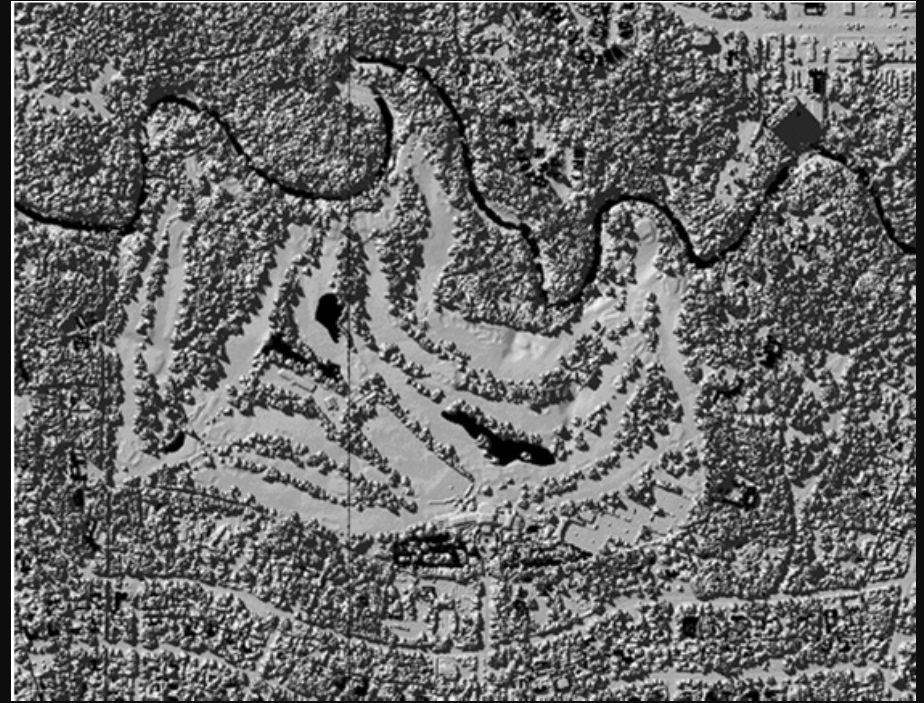
Digital Surface Model (DSM)

DSM

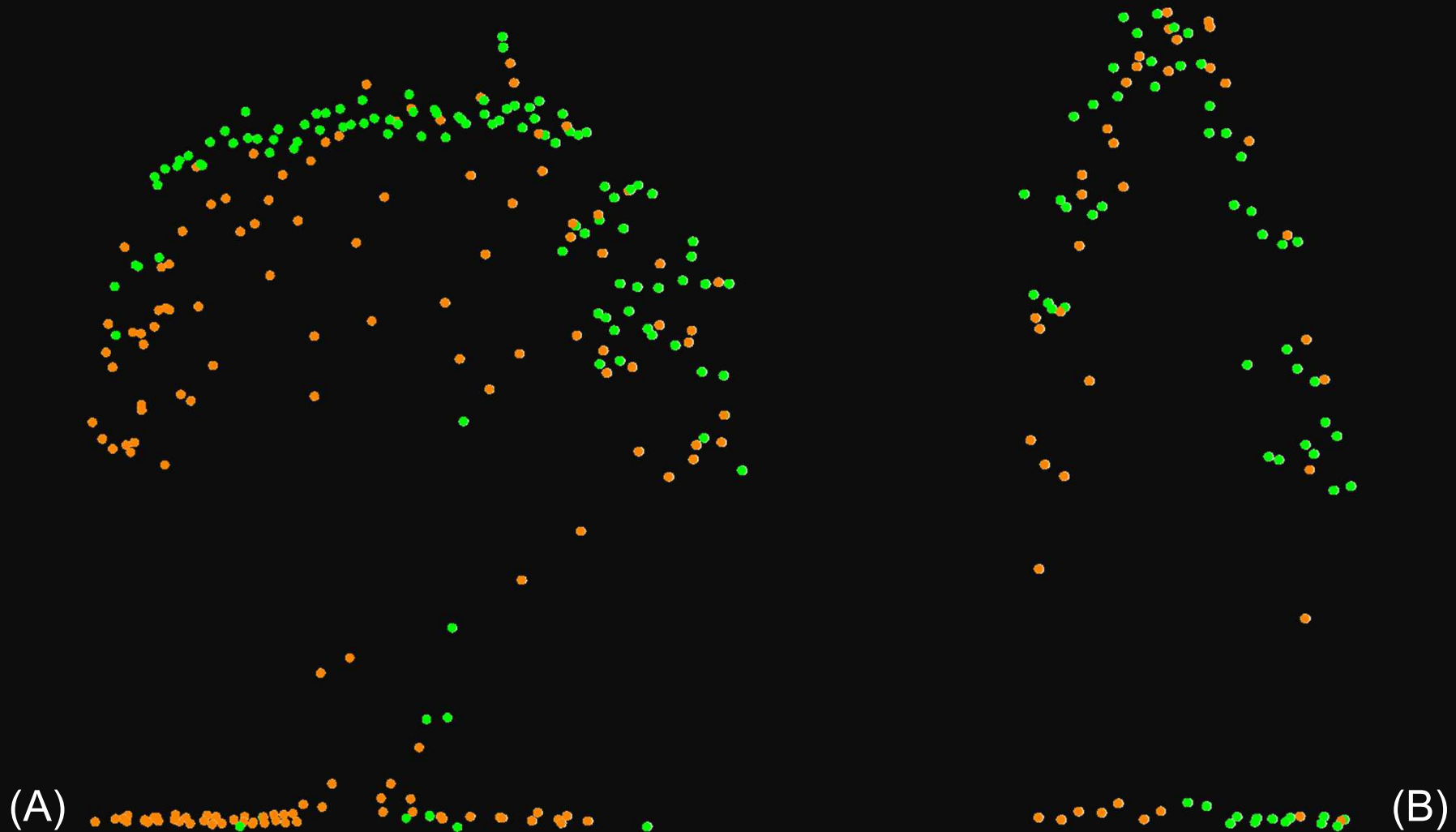
Raw LiDAR signals are reflected from trees, buildings, and the grounds

Interpolation results in a digital surface model

Vegetation canopy model, 3D urban models

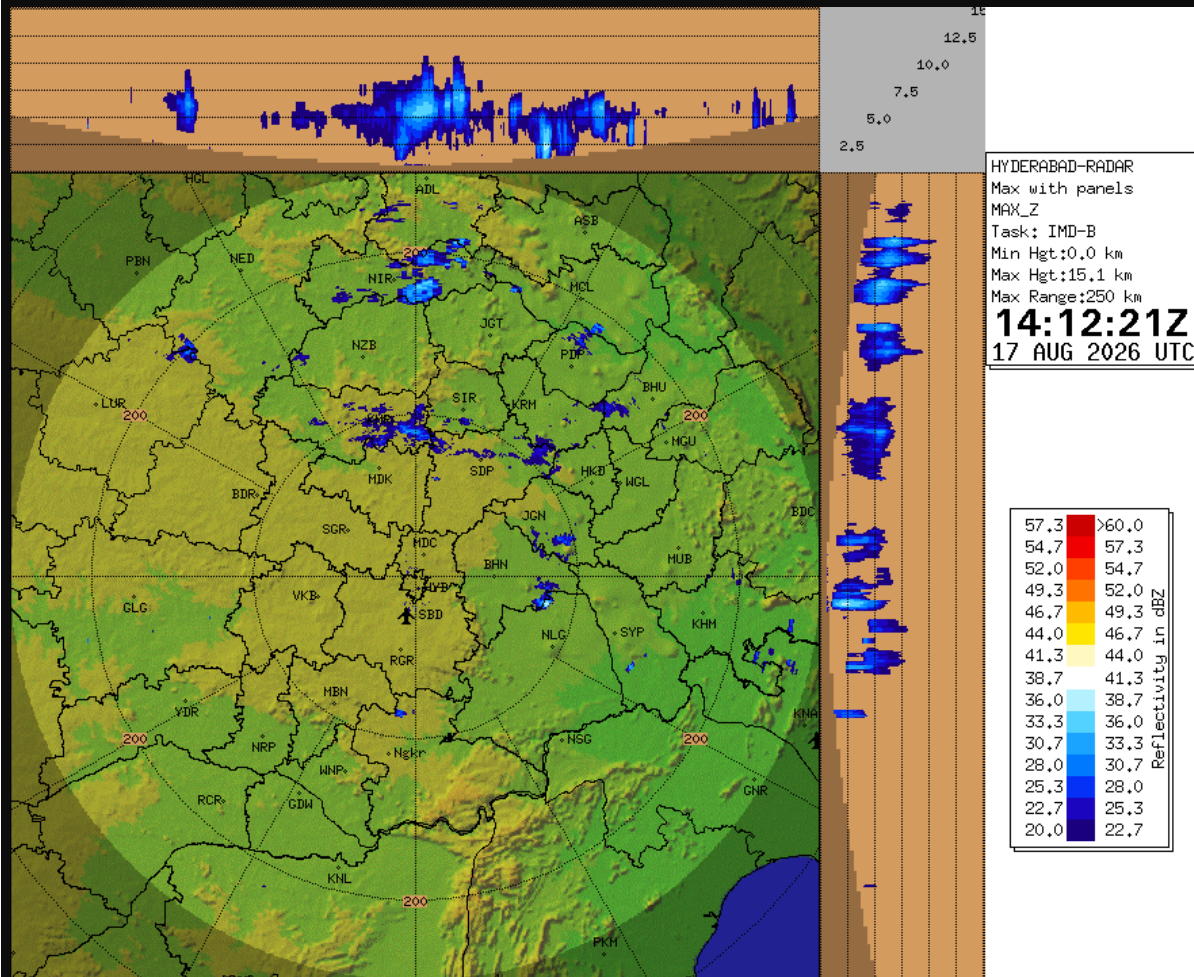


Leaf-on vs. leaf-off



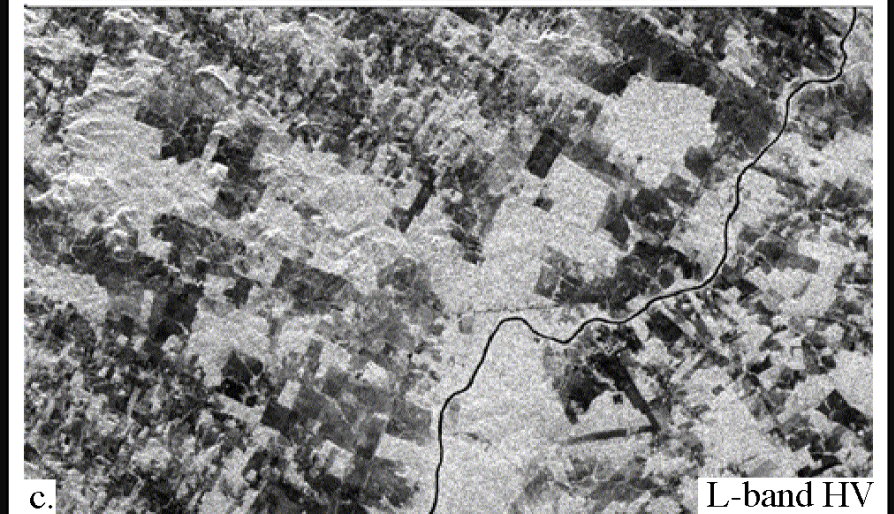
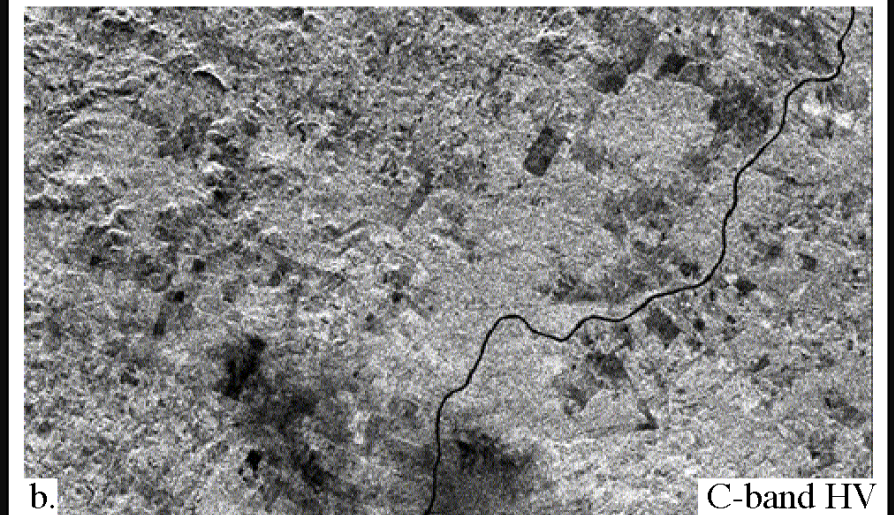
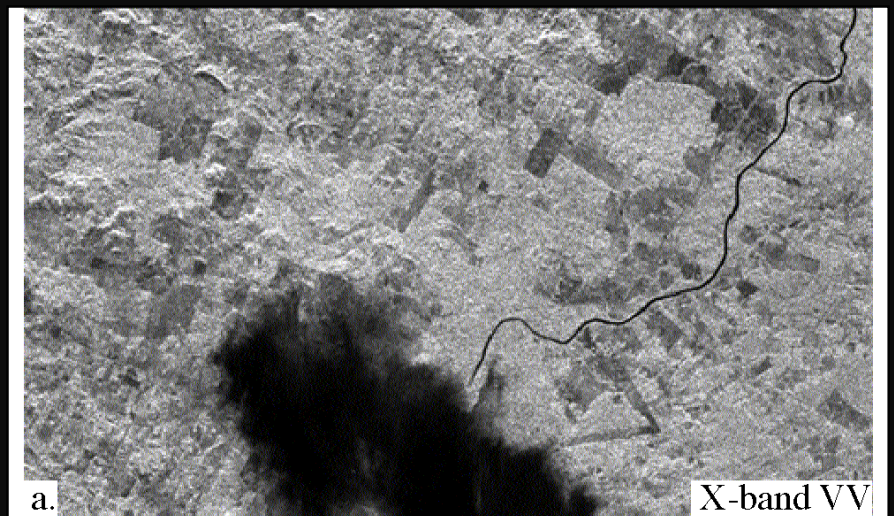
Cross section of LIDAR data through a single deciduous tree (A) and coniferous tree (B) including bare-earth returns. The green dots represent leaf-on returns and the brown dots represent leaf-off returns

RADAR / SAR

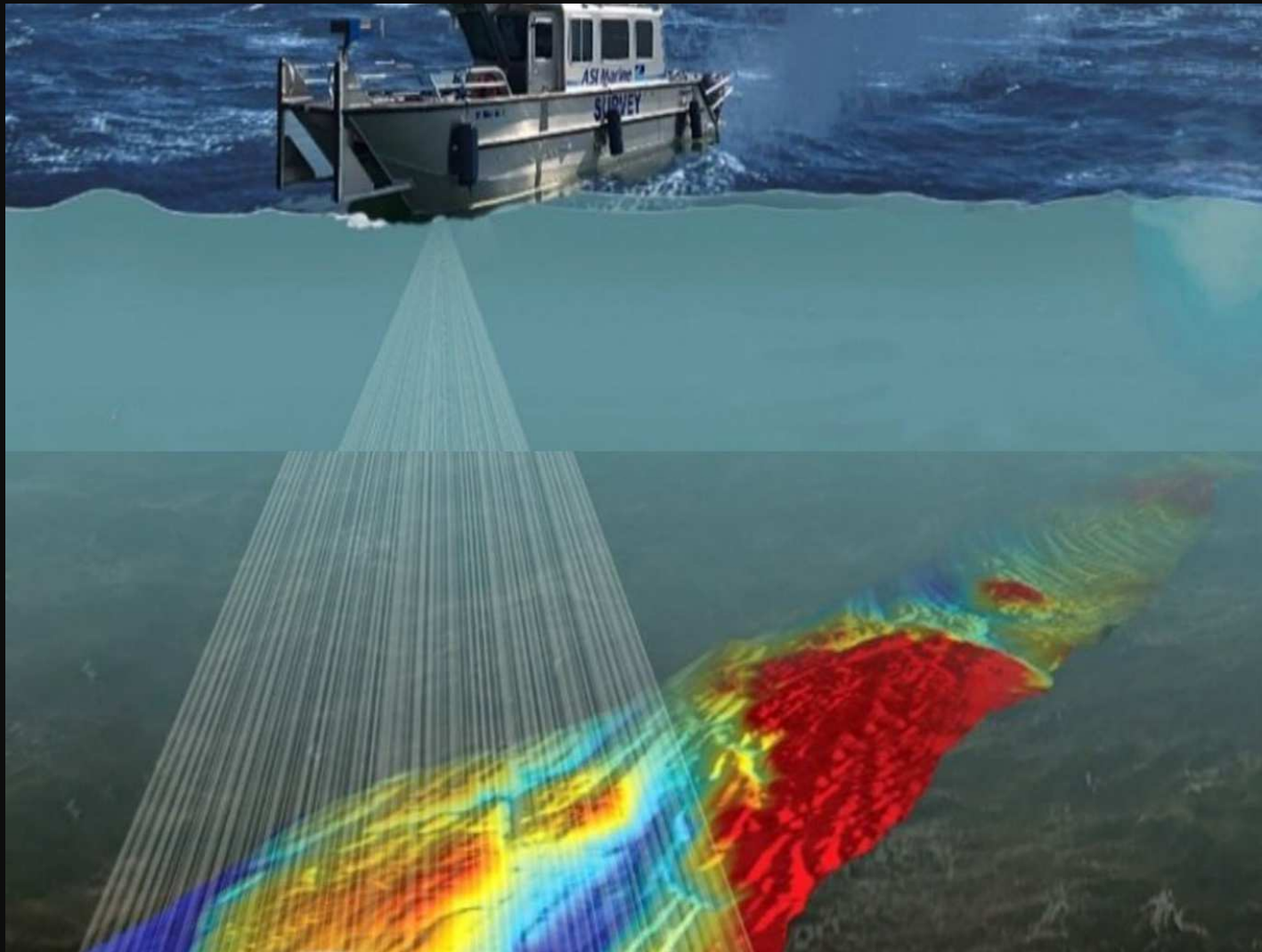


DOPPLER WEATHER RADAR IMAGES

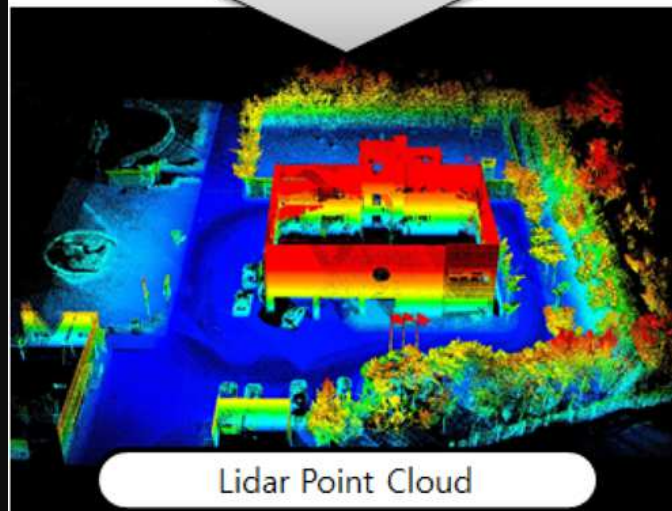
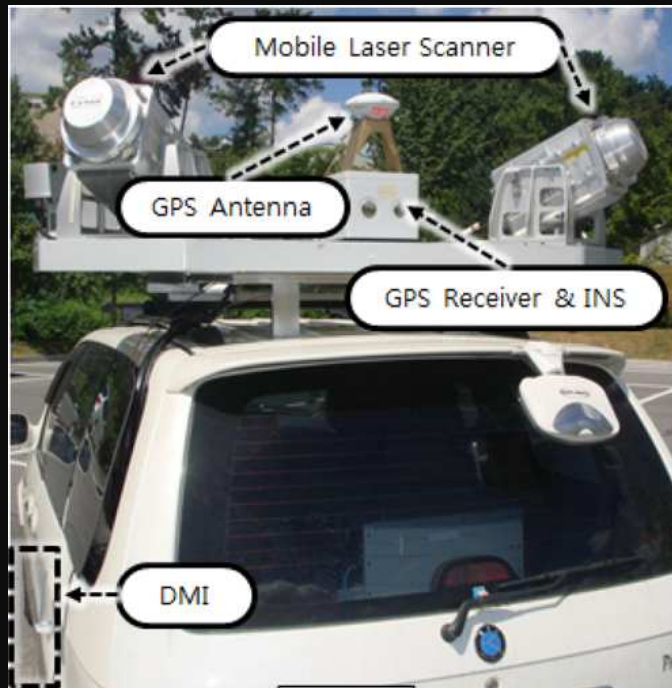
SIR-C/X-SAR Images of a Portion of
Rondonia, Brazil, Obtained on April 10,
1994



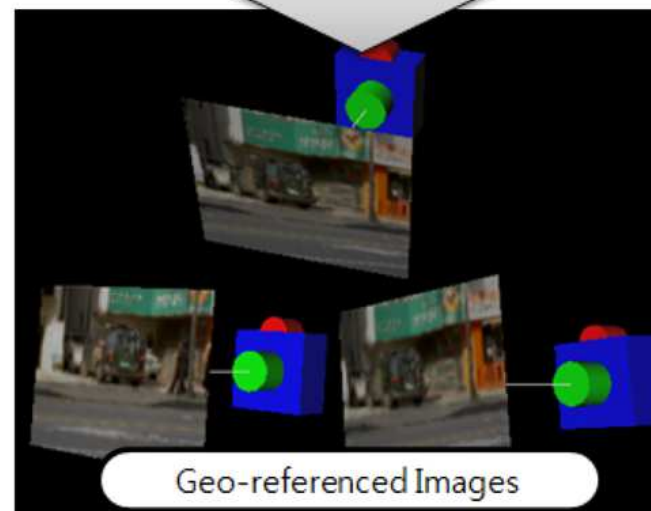
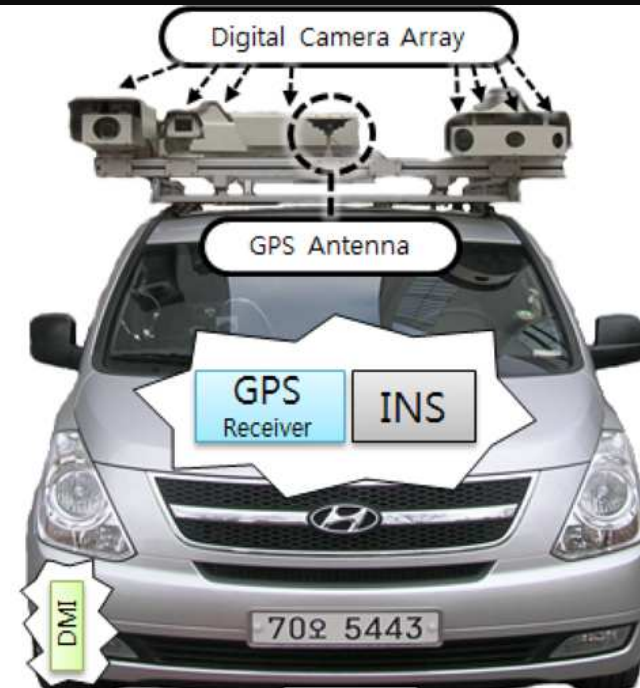
Sonar/Bathymetry



Digital Field Cameras & Mobile Mapping



(a) Mobile Laser Scanner
Centered LMMS (LMMS 1)

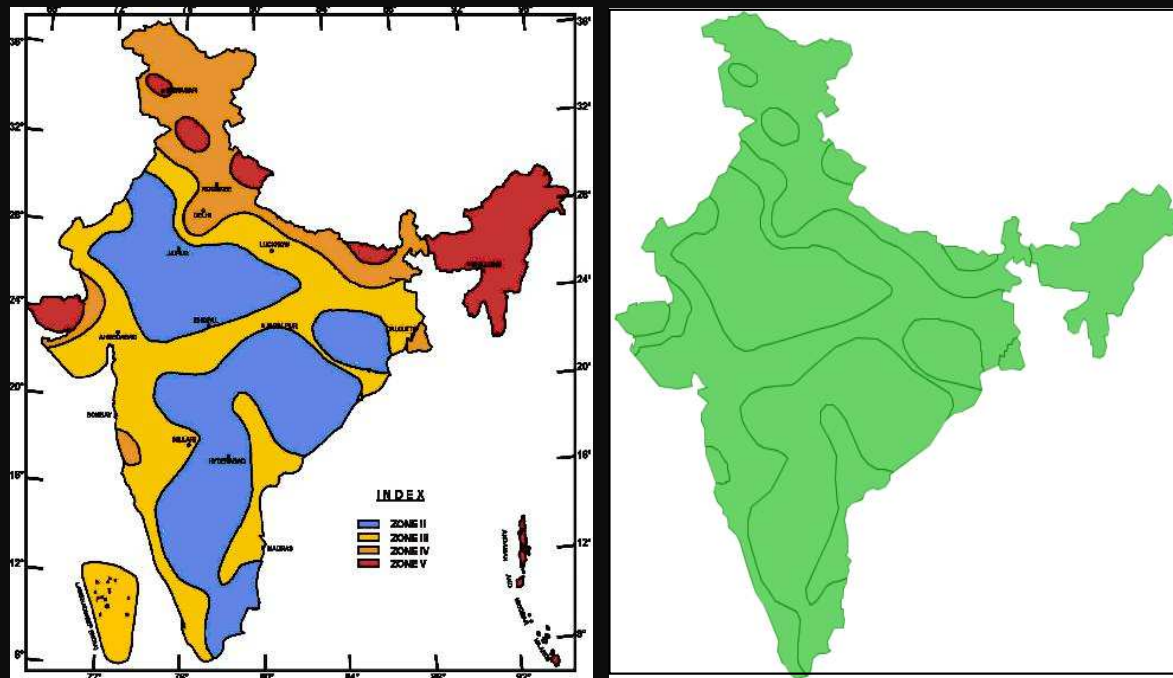


(b) Digital Camera
Centered LMMS (LMMS 2)

Secondary Raster data capture

1. Digital Scanning

Scanned maps and documents are used extensively in GIS as background maps and data stores.

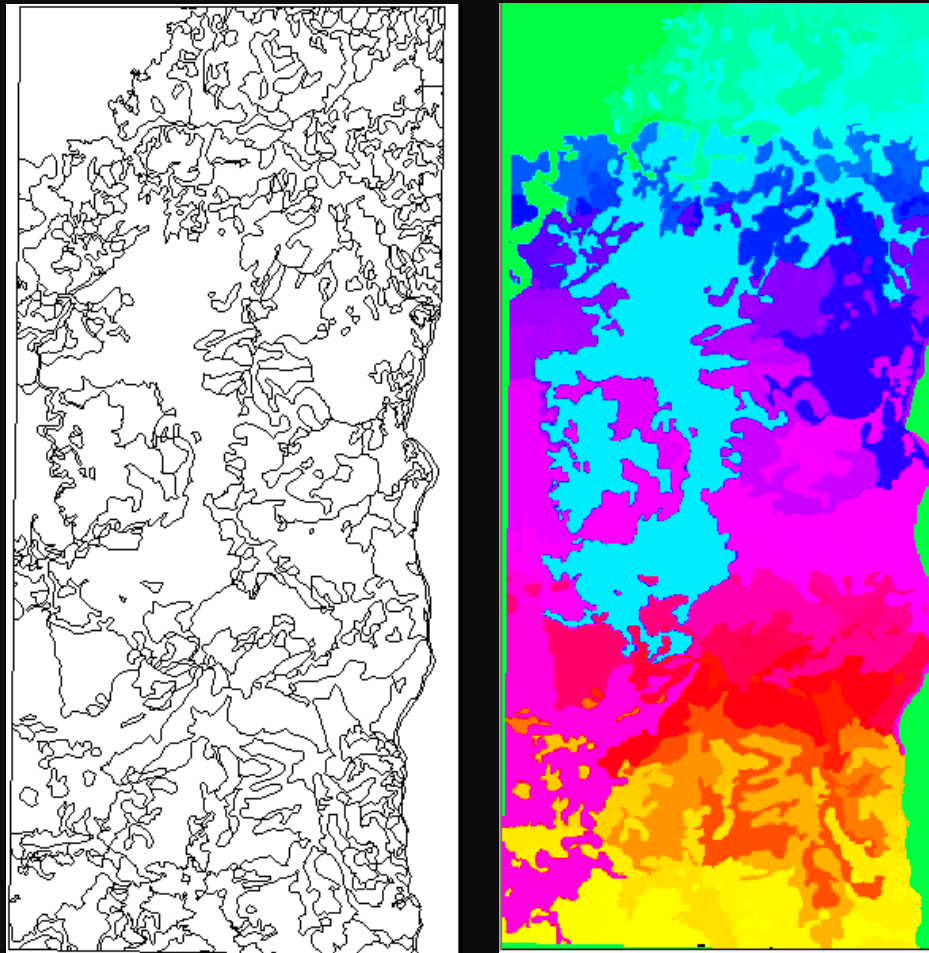


Seismic zones of India

Quality of outputs
Source map quality
Scanning process

2. Rasterization

Rasterization refers to conversion from vector to raster data.

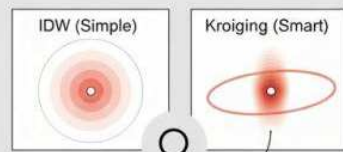
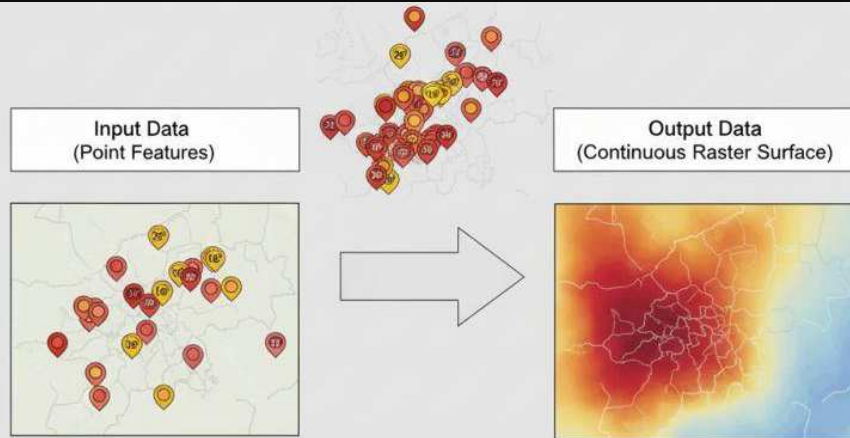
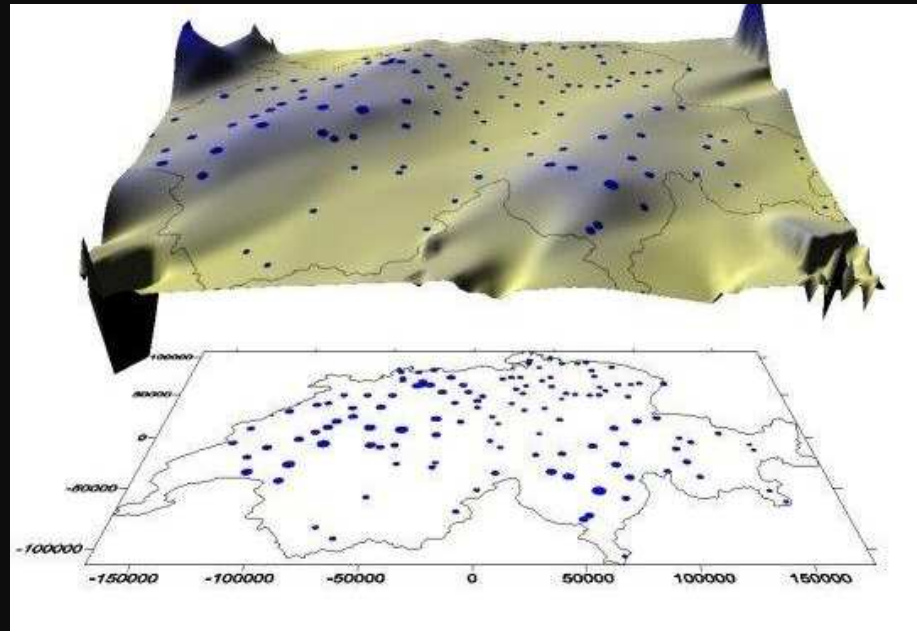
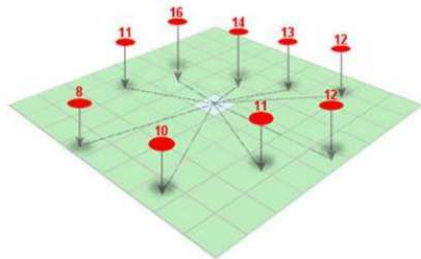


Forest type map of North Andaman

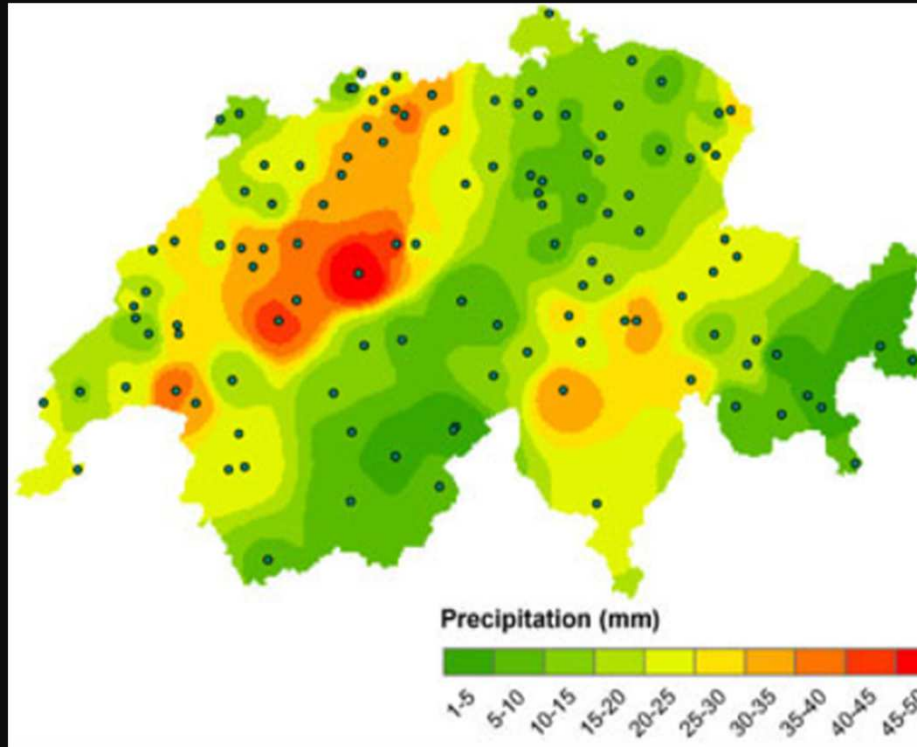
3. Spatial interpolation

Interpolation

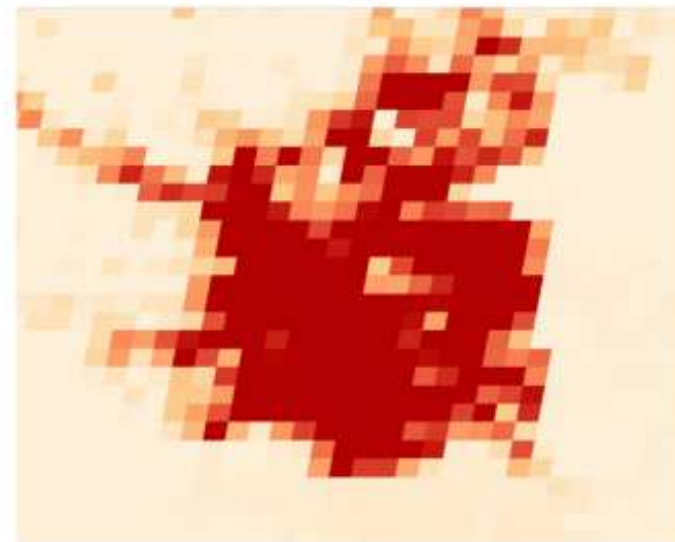
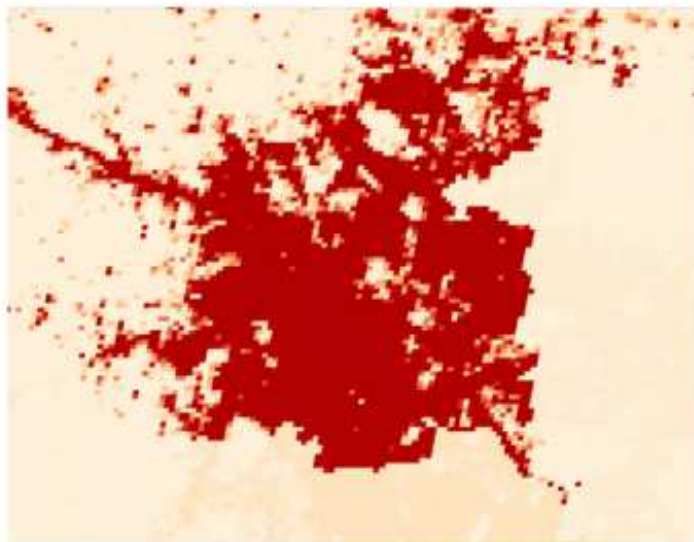
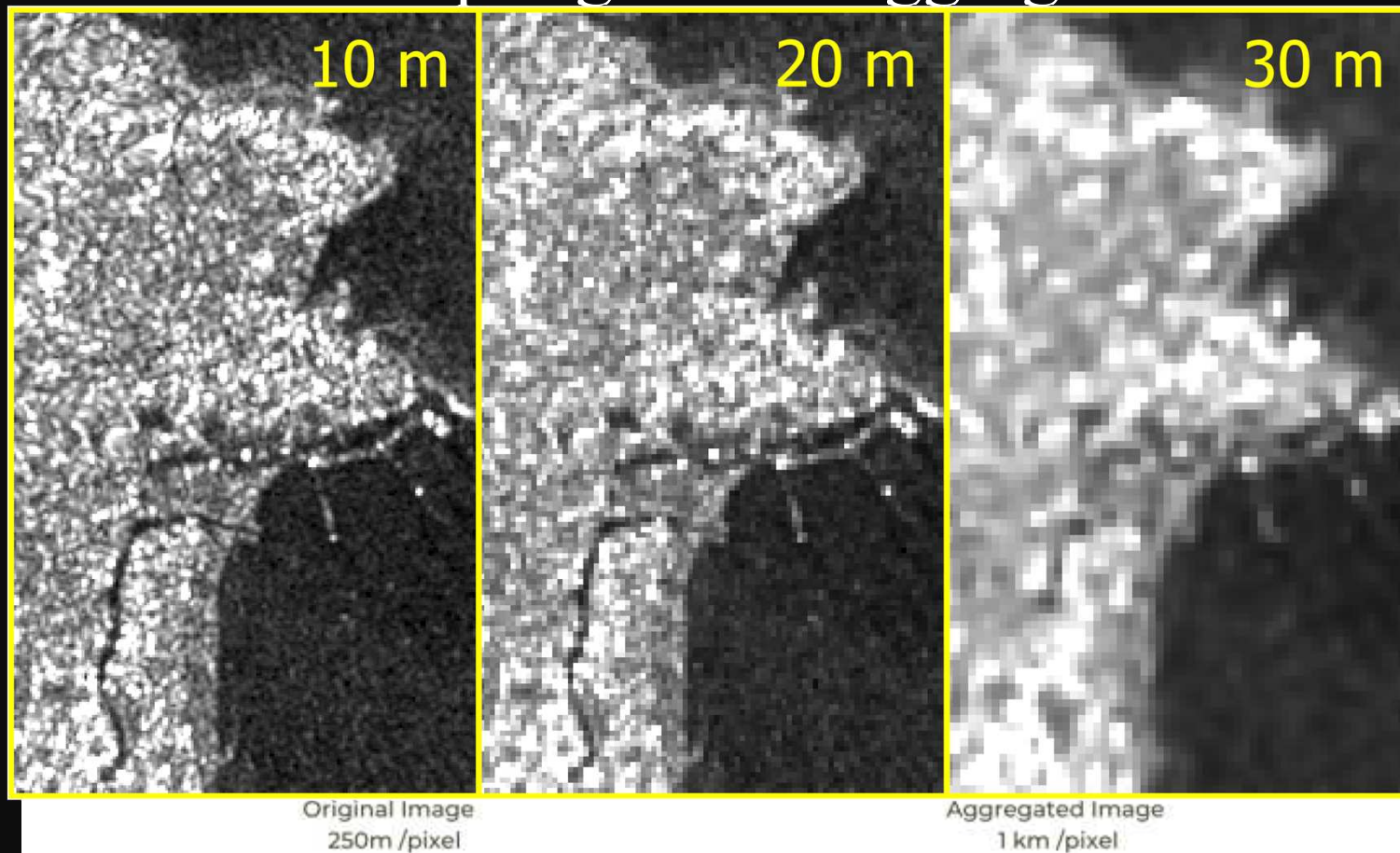
The unknown value of the cell is based on the values of the sample points as well as the cell's relative distance from those sample points.



Predicting Unknown Values Based on Distance & Statistics



4. Resampling / Grid Aggregation:



3. Data Transfer

What types of raster data can be transferred?

Raster data	Example
Satellite imagery	Landsat, Sentinel-2
DEM/DTM	Elevation raster
Aerial imagery	Orthophoto
Climate surfaces	Temperature, rainfall
Thematic rasters	Land use, soil, population density
Scanned maps	Digital TIFF/JPEG map

Formats

What to check ?